

KHADIJA BATOOL (Ph.D., M.Sc., Green Card Pending)

Boyce Thompson Institute at Cornell University
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Summary of qualifications

- Micro/Molecular biologist with a keen interest in molecular mechanisms of pathology
- Extensive experience in Host-pathogen interaction research using molecular, cell culture and in vivo techniques
- Extensive experience in developing molecular, cell and virus based diagnostic assays
- Skilled bench scientist with a strong publication, mentoring and safety officer history

Experience

Boyce Thompson Institute at Cornell University (Postdoctoral scientist) (2022 – present)

- Host-Virus interaction: why are baculovirus infections not lethal to some lepidopteran species?
- Formulated and executed a project to analyze baculovirus gene expression in permissive and semi-permissive insect cell lines using Comparative Transcriptome analysis to understand the dynamics of viral expression.
- Submitted two first authored research articles.

Southern Medical University (China), Division of Biology (Postdoctoral scientist) (2019 – 2022)

- Assessed the potential of recombinant mosquito densovirus as a biological control agent against the *Ae. albopictus* mosquito.
- Developed a series of novel larvicide oil suspensions (denoted *Bacillus thuringiensis* (Bti) oil, *Ae. albopictus* densovirus (AalDV-5) oil, and a mixture of AalDV-5+Bti oil), which were tested against *Ae. albopictus* larvae under experimental semi-field and open-field conditions.
- Transfected *Ae. albopictus* C6/36 cell lines with recombinant densoviruses to express viral proteins.
- Published two first authored research articles and won funding grant for the project.

Fujian Agriculture and Forestry University, Dept. of Biochemistry & Molecular biology (PhD student) (2015 – 2019)

- Transcriptome analysis of *Aedes aegypti* midgut in response to mosquitocidal *Bacillus thuringiensis* LLP29 toxin.
- Identified immune genes in the midgut and biochemical pathways induced in insects in response to toxin infection using bioinformatics tools.
- Characterized differentially expressed midgut CTL genes and study their interactions with mosquito receptors.
- Toxic Activity, Molecular Modeling and Docking Simulations of *Aedes aegypti* CTL variant and study their interaction with APN receptor.
- Published three primary-authored and two co-authored research articles.

Education

Ph.D. Dept. of Biochemistry & Molecular biology, Fujian Agriculture and Forestry University (2019)

M.Sc. Dept. of biology, Arid Agriculture University, Pakistan (2014)

B.Sc. (Major in Biology), Univ. of Punjab, Pakistan (2010)

Leadership and scientific activities

Management

- Trained 2 undergraduate students and junior personnel from diverse backgrounds.
- Part-time Research Assistant at Key Laboratory of Biopesticide and Chemical Biology. Conducted research on Insect biocontrol.
- Maintained and ordered laboratory supplies for over 2 years.
- Laboratory safety officer for 2 years

Community Education and Outreach

- Attended 13th National Symposium on Bacillus Youth Workers, Tianjin Society of Microbiology, Nov-9 to Nov-11, 2018, Nankai University, China.
- Attended 12th National Symposium on Bacillus Youth Workers from Dec-1 to 2017-Dec-3 at College of Life Science, Nanjing Agriculture and Forestry University, China.
- Review of Scientific Manuscripts: Microorganisms, Agrochemicals, Genes, Experimental Parasitology, and Parasites & Vectors.

Awards

- Participated in “Winter Symposium of College of Life Science” held at Fujian Agriculture and Forestry University, January 2019, and awarded First position.
- Awarded Excellent Academic Report at Nankai University (2018).
- Awarded with Fujian Agriculture and Forestry University, China, scholarship for the Ph.D. degree (2015-2019).
- Granted the Fellowship of China Postdoctoral Science Foundation funding (2020M672715).

Publications (selected list from 24)

Khadija Batool, Jeffrey Hodgson, Shan Wu, Zhangjun Fei, Gary W. Blissard. Transcriptome analysis of baculovirus AcMNPV infections of permissive and semi permissive cell lines. (2025) (preparing for journal).

Lu Yang, Yonghui Gao, Yulan Chen, Peiwen Liu, **Khadija Batool**, Jianxia Tang, Jinbao Gu. AalpiRNA-18529 regulates vitellogenesis of *Aedes albopictus* via Gadd45a-mediated JNK-dependent nurse cell apoptosis pathway, *elife*, (2025) (under review).

Surayya Lawan Idris, Aisha Lawan Idris, **Khadija Batool**. Bacilli: The keystone biocontrol agents eradicating phytopathogens, *Research & Reviews: Journal of Microbiology & Virology* 13 (2023).

Khadija Batool, Jie Xiao, Siyu Zhao, Yu-Gu Xie, Jinbao Gu, Xiao-Guang Chen. Densovirus oil suspension significantly improves the efficacy and duration of killing the vector mosquito, *Aedes albopictus*, *Viruses* 14, (2022).

Khadija Batool, Intikhab Alam, Peiwen Liu, Zeng Shu, Siyu Zhao, Wenqiang Yang, Xiao Jie, Jinbao Gu, Xiao-Guang Chen. Recombinant Mosquito Densovirus with Bti toxins significantly improves pathogenicity against *Aedes albopictus*, *Toxins* 14, (2022).

Intikhab Alam, **Khadija Batool**, Aisha lawan Idris, Lingling Zhang. Role of Lectin in response of *Aedes aegypti* against Bt Toxin, *Frontiers in Immunology* 10, (2022). **(Equal Contribution)**.

Intikhab Alam, **Khadija Batool**, Aisha lawan Idris, Xiong guan and Lingling Zhang. Function of residue Leucine-6 in Modulation Cry11Aa toxicity in *Aedes aegypti*, *Frontiers in Immunology* 13, (2022). **(Equal Contribution)**.

Khadija Batool, Intikhab Alam, Liang jin, Jin Xu, Junxiang wang, Enjiong Huang, Xiong Guan, Xiao-Qiang Yu and Lingling Zhang. CTLGA9 interacts with ALP1 and APN receptor to modulate Cry11Aa toxicity in *Aedes aegypti*. *Journal of Agriculture and Food Chemistry* 67, (2019).

Khadija Batool, Intikhab Alam, Songqing Wu, Wencheng Liu, Guohui and Lingling Zhang. Transcriptomic analysis of *Aedes aegypti* in response to Mosquitocidal *Bacillus thuringiensis* LLP29 toxin, *Scientific reports* 8, (2018).

Khadija Batool, Intikhab Alam, Guohui Zhao, Junxiang Wang, Jin Xu, Xiaoqiang Yu, Enjiong Huang, Xiong Guan and Lingling Zhang. C-Type Lectin-20 Interacts with ALP1 receptor to reduce Cry toxicity in *Aedes aegypti*, *Toxins* 10, (2018).

Jin Xu, Chenxu Wu, Zhaohui Yang, Wencheng Liu, Hong Chen, **Khadija Batool**, Junmin Yao, Xiao Fan, Lei Xu, Xiong Guan, Lingling. RecG is involved with the resistance of Bt to UV. *Pesticide biochemistry and physiology* 167, (2020).

Xiaohua Hu, Hong Chen, Jin Xu, Guohui Zhao, Xianhui Huang, Jiannan Liu, **Khadija Batool**, Chenxu Wu, Songqing Wu, Enjiong Huang, Juan Wu, Munmun Chowhury, Jie Zhang, Xiong Guan, Xiao-Qiang Yu, Lingling Zhang. Function of *Aedes aegypti* galectin-6 in modulation of Cry11Aa toxicity, Pesticide biochemistry and physiology 162, (2019).

Guihui Zhao, Liu Jinan, Hu Xiaohua, **Khadija Batool**, Jin Liang, Xiaoqiang Yu, Enjiong Huang , Xiong Guan and Lingling Zhang, Cloning, expression and activity of ATP-binding protein in *Bacillus thuringiensis* toxicity modulation against *Aedes aegypti*, Parasites & Vectors 12, (2019).

Ling-Ling Zhang, Xiao-Hua Hu, Song-Qing Wu, **Khadija Batool**, Munmun Chowdhury, Yi Lin, Jie Zhang, Sarjeet S Gill, Xiong Guan, Xiao-Qiang Yu. *Aedes aegypti* galectin competes with Cry11Aa for binding to ALP1 to modulate Cry toxicity, Journal of Agriculture and Food Chemistry 66, (2018).

Ling-Ling Zhang, Zhang XiaoJuan Zhang XiaoJuan, **Khadija Batool**, Hu XiaoHua, Chen MingFeng, Xu Jin, Wang JunXiang, Pan XiaoHong, Huang TianPei, Xu Lei Xu Lei, Yu XiaoQiang, Guan Xiong. Comparison and Mechanism of the UV-Resistant Mosquitocidal Bt Mutant LLP29-M19, Journal of Medical Entomology 55 (2018).

Junxiang Wang, Guifang Lin, **Khadija Batool**, Shuaiqi Zhang, Mingfeng Chen, Jin Xu, Juan Wu, Liang Jin, Ivan Gelbič, Lei Xu, Lingling Zhang, Xiong Guan. Alimentary Tract Transcriptome Analysis of the Tea Geometrid, *Ectropis oblique* (Lepidoptera: Geometridae), Journal of Economic Entomology 111, (2018).

Lingling Zhang, Guohui Zhao, Xiaohua Hu, Jiannan Liu, Mingwei Li, **Khadija Batool**, Mingfeng Chen, Junxiang Wang, Jin Xu, Tianpei Huang, Xiaohong Pan, Lei Xu, Xiao-Qiang Yu, Xiong Guan. Cry11Aa Interacts with the ATP-Binding Protein from *Culex quinquefasciatus* To Improve the Toxicity, Journal of Agricultural and Food Chemistry, 65, (2017).